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PHNOM PENH INTERNATIONAL UNIVERSITY

Car Price Prediction Using Machine Learning

LINEAR REGRESSION × XGBOOST

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What we'll Cover



1. Introduction



2. Dataset Description



3. Methodology



4. Compare Model



5. Conclusion

1. Introduction

Why a data-driven approach matters

PROBLEM STATEMENT

Accurate pricing is essential for buying and selling used cars. Manual methods are often inconsistent and inaccurate.

OBJECTIVE

Build price-prediction models, compare Linear Regression with XGBoost, and identify the stronger algorithm for this task.

1.1 Algorithm 1: XGBoost

Gradient boosting for complex, nonlinear relationships

WHY WE CHOOSE THIS MODEL?

High predictive accuracy on complex tabular data, with the ability to capture nonlinear feature interactions.

ADVANTAGES

- Strong predictive accuracy
- Handles nonlinear patterns
- Regularization reduces overfitting

LIMITATIONS

- More computationally intensive
- Multiple hyperparameters require tuning

1.2 Algorithm 2: Linear Regression

A transparent baseline for price prediction

WHY WE CHOOSE THIS MODEL?

Simple, interpretable, and fast to train — an effective benchmark for more complex models.

ADVANTAGES

- Fast computation
- Simple and interpretable

LIMITATIONS

- Assumes linear relationships
- Sensitive to outliers

2. Dataset Description

CarDekho Used Car Dataset

8,128

records

12

features

selling_price

target variable

KEY FEATURES

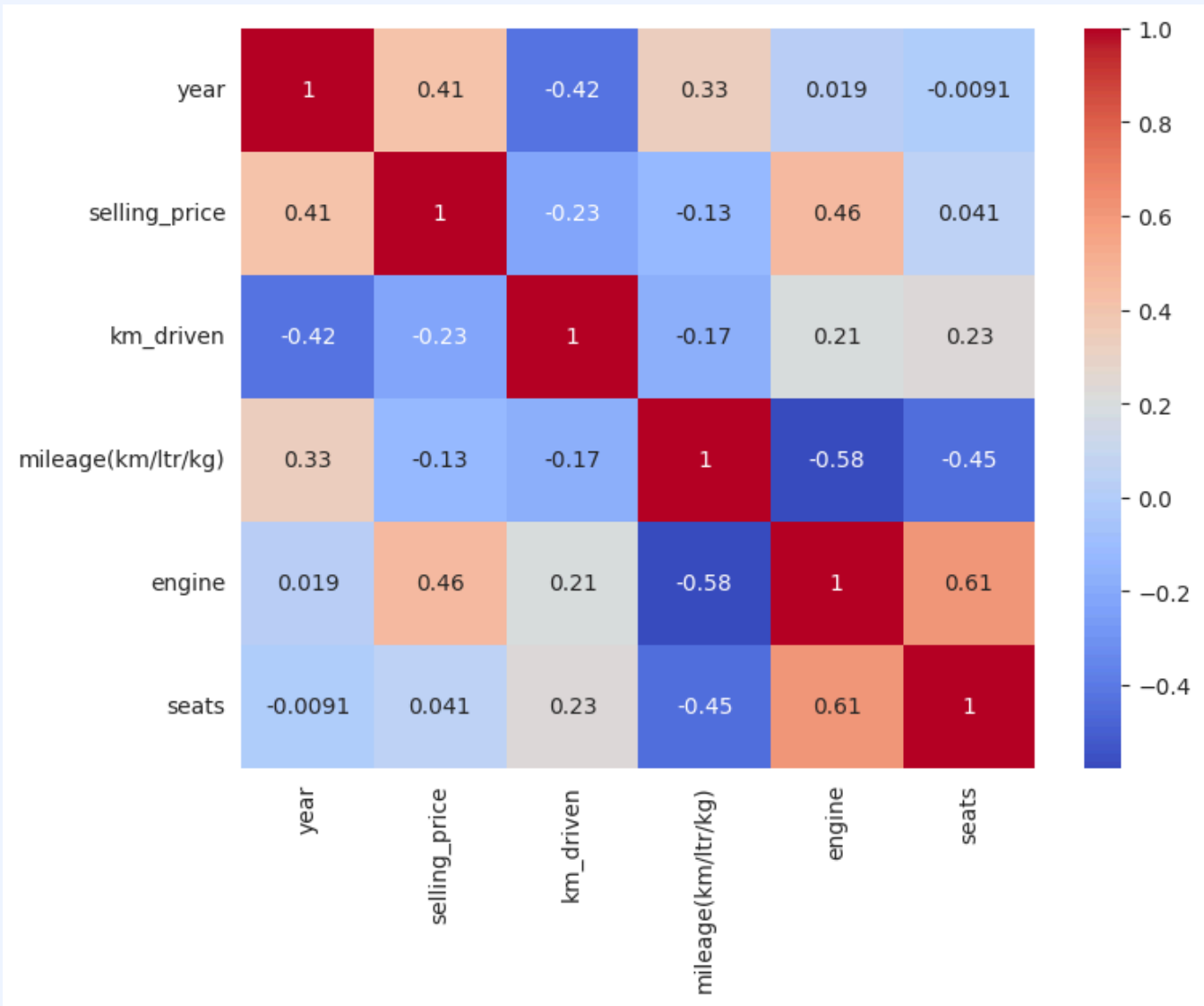
Year • Fuel Type • Transmission • Engine • Mileage • Seats
Owner • Seller Type • Max Power

A credible, domain-relevant used-car dataset suited to supervised price modeling.

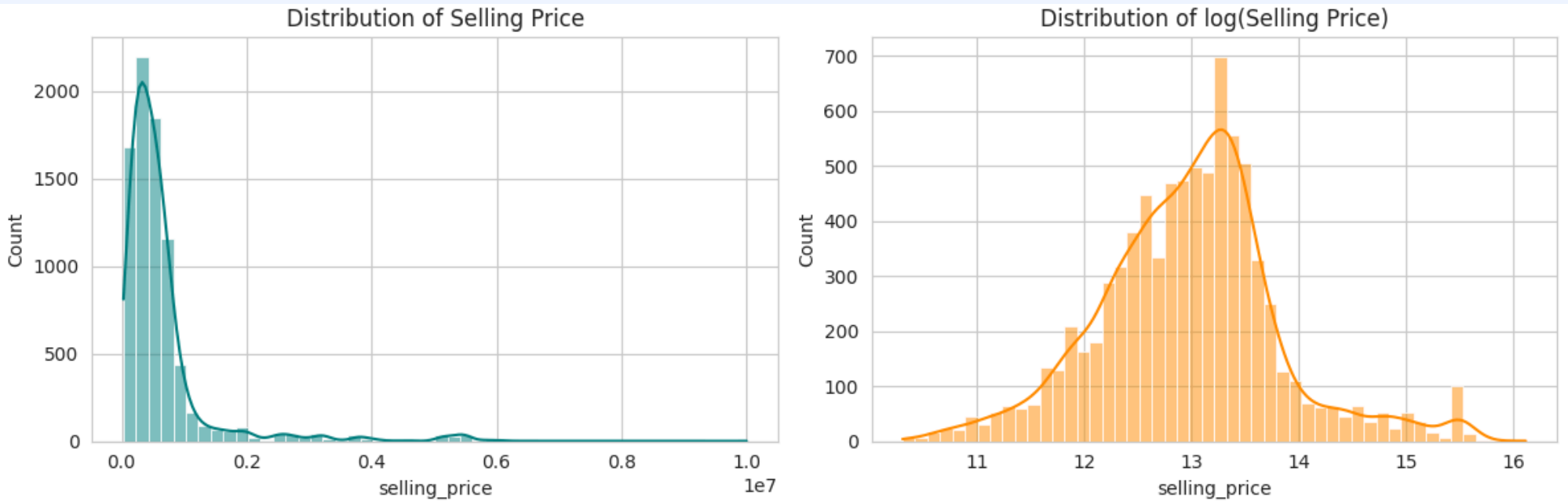
2.1 Data Insights

Exploratory patterns in the market

Correlation Heatmap



Selling price: Raw vs Log_tranform



2.2 Data Preprocessing

A consistent pipeline from raw records to model-ready features

1

Remove duplicates

2

Fill missing values

3

Convert data types

4

Extract car brand

5

Encode categories

6

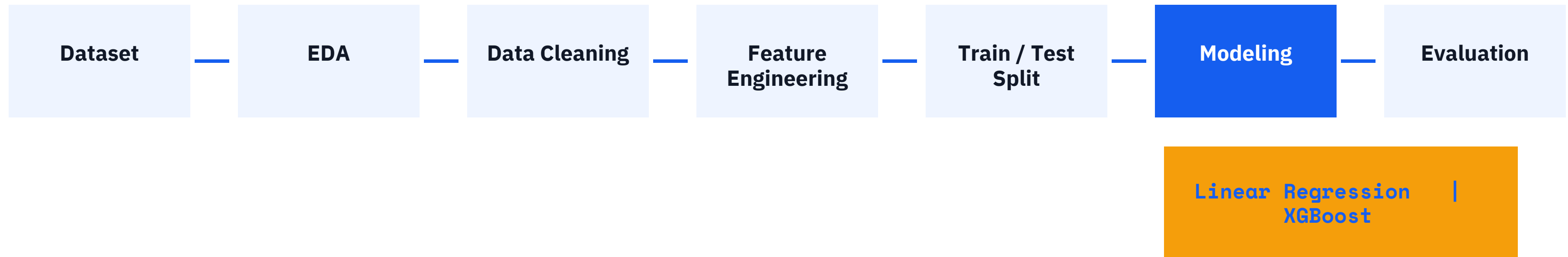
Train / test split

7

Scale for Model

3. Methodology

End-to-end modeling workflow



Comparative Analysis

Use test-set metrics from the final model run

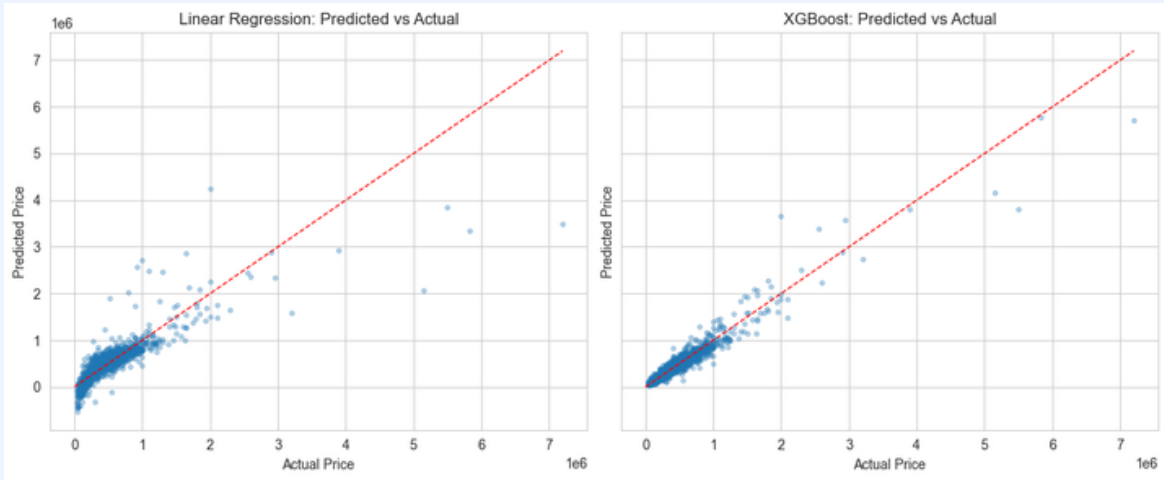
METRIC	LINEAR REGRESSION	XGBOOST	BETTER
MAE	−132065.228197	−75538.945312	XGBOOST
MSE	−6.406271e+10	−132629.558093	XGBOOST
RMSE	−253106.121714	−123,347	XGBOOST
R² SCORE	−0.707905	−0.919795	XGBOOST

Decision rule: lower error metrics and a higher R² indicate the stronger fit. XGBoost is expected to benefit from nonlinear modeling and regularization; confirm with measured results.

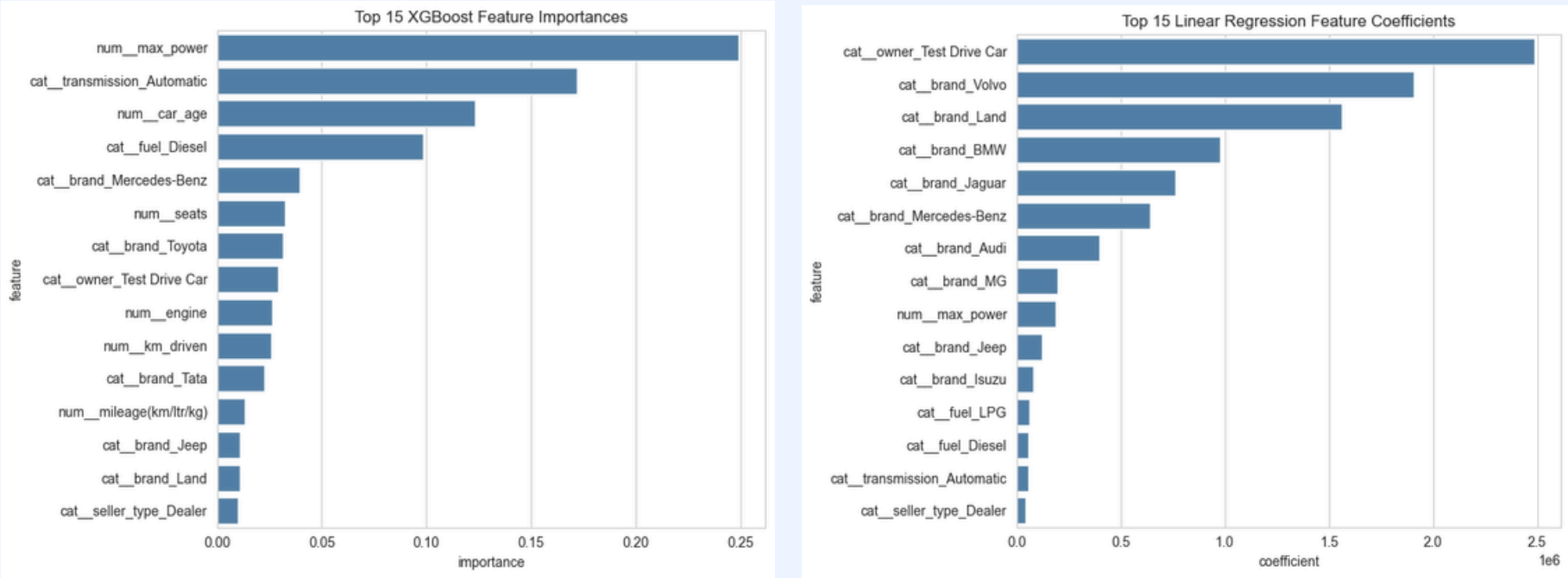
Visual Comparison

Actual vs. predicted selling prices

Actual price vs Predict



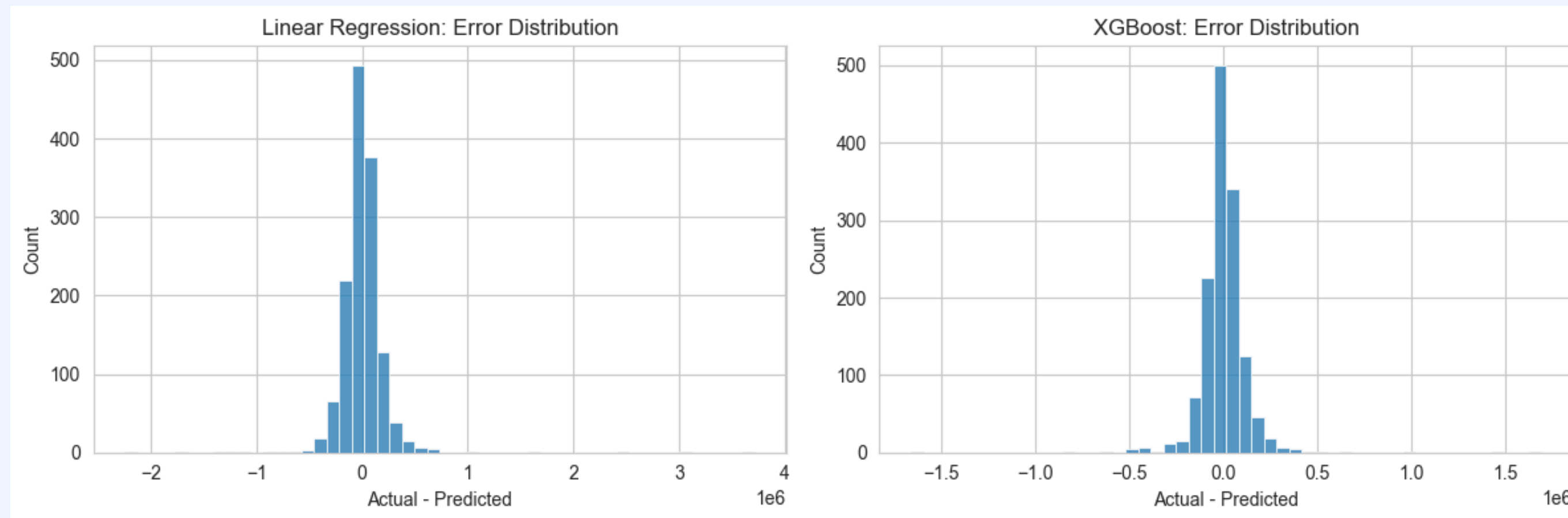
Feature Importances



Visual Comparison

Actual vs. predicted selling prices

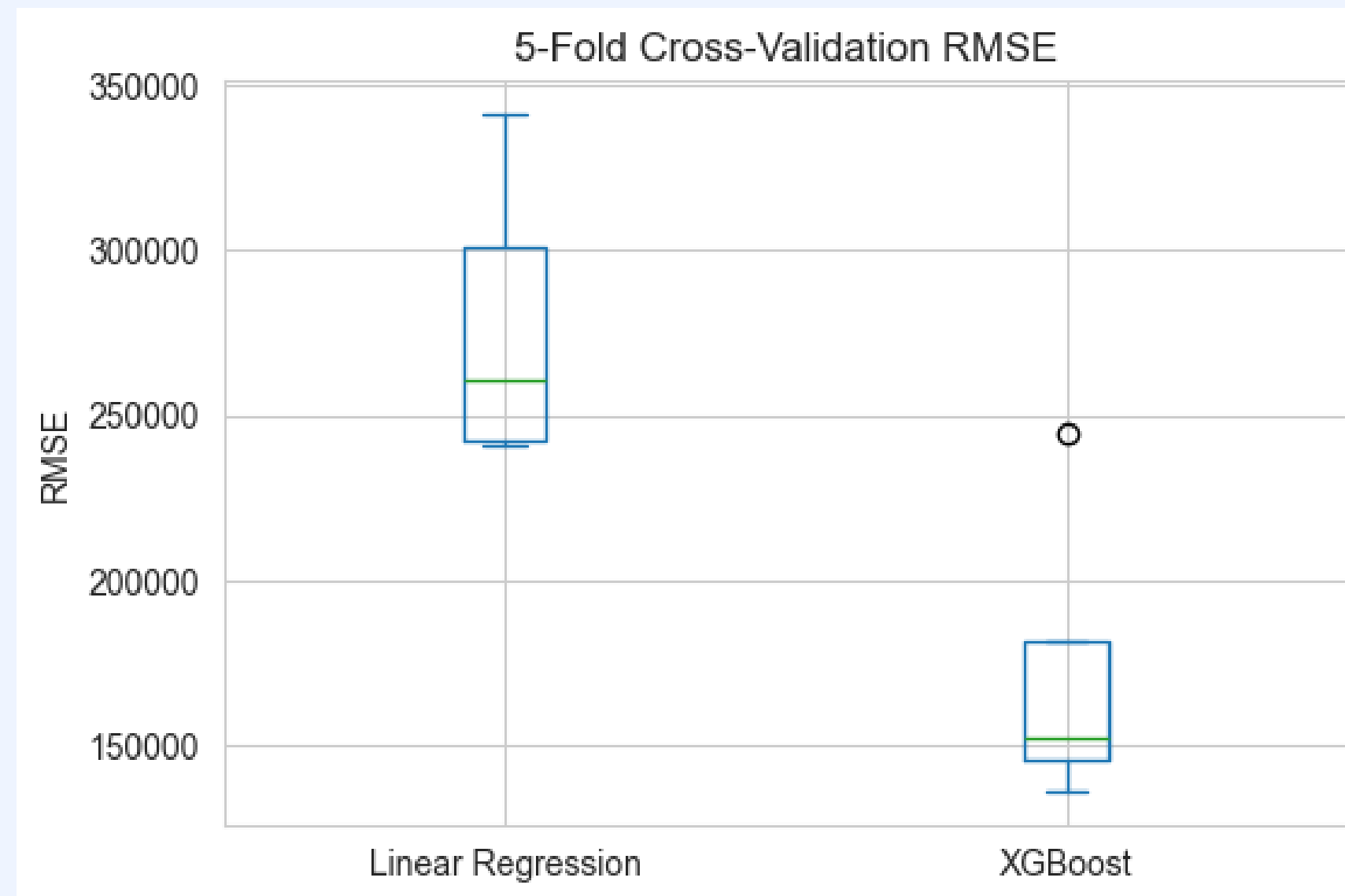
Residual (Error)



Visual Comparison

Actual vs. predicted selling prices

5 Fold Cross-Validation RMSE



Conclusion

What the comparison tells us

SUMMARY

Both models were built and compared. XGBoost showed stronger overall performance; Linear Regression remained a useful, interpretable baseline.

CHALLENGES

- Missing values and duplicates
- Effective categorical encoding
- XGBoost hyperparameter selection

References

Sources and technical documentation

01 CarDekho Used Car Dataset

02 Scikit-learn Documentation — scikit-learn.org

03 XGBoost Documentation — xgboost.readthedocs.io

Thank You